TRAG

Risk screening of waste treatment facilities using mass balance, process models and independent data sources

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1. Summary

TRAG is a research project at the University of Zagreb. Its aim is to check whether the waste quantities that treatment facilities report each year are physically possible, and to rank facilities by how far their reports deviate from what their process and capacity allow.

The basic idea is a mass balance for every facility and every year. What enters a plant has to leave it as waste, as product, as a change in stock, or as a known process loss. Where the reported numbers do not close, the facility is flagged for closer inspection. The result is a list of deviations to be verified, not a finding of wrongdoing.

A first analysis of the public Croatian Environmental Pollution Register (ROO) shows that the data needed for this exists. The facility in Gospić that is now under criminal investigation reported about 15,600 t of treated waste for 2023 and only 19 t of residual waste, and filed no reports after that.

2. Problem

Waste fraud usually follows a simple pattern. A company is paid to accept waste, reports it as recycled or as secondary raw material, and then stores, buries or dumps it. When the company stops operating, the cost of clean-up falls on the public. EU estimates put the share of illegal waste shipments at 15–30%, with a value of up to EUR 9.5 billion per year.

In Croatia, a licensed facility in Gospić received waste from Italy, Slovenia and Croatia that was declared as recyclable plastic. Several thousand tonnes accumulated on site and part of it was moved to an unlicensed location. A criminal investigation started in February 2025.

Facilities already report detailed annual data to the register. These reports are collected and published, but they are not checked for physical consistency, for example whether a plant with a given process and capacity can treat 15,000 t of waste and produce almost no residue.

3. Approach

For each facility and year, TRAG starts from the balance

$$\text{input} = \text{output} + \Delta\text{stock} + \text{process losses}$$

and adds four checks:

1. **Mass balance** per facility, using data reconciliation with uncertainty ranges and gross error tests.
2. **Cross-checks between partners**: quantities that facility A reports as sent compared with what facility B reports as received.
3. **Physical feasibility**: permitted capacity, storage area, material density and equipment throughput.
4. **Chemical feasibility**: for example, hazardous waste entering and non-hazardous waste leaving a plant whose permitted process cannot achieve this.

Because reported data can be made internally consistent, the checks are compared with sources that are harder to falsify: Sentinel-2 satellite images of stockpile area over time, published financial statements (whether the claimed recycled product appears in sales), permit history and company records.

The work is organised in four steps of increasing cost:

1. Annual screening of all treatment sites in Croatia (about 450) with a set of indicators.
2. Process models for the main plant types, giving the expected share of residue with an uncertainty range. This turns the input/output ratio into a statistical test.
3. Satellite time series for the highest-ranked sites.
4. Field checks with a drone for selected sites.

For every flagged facility, the output states which check failed and by how much.

4. Preliminary results

ROO publishes facility-level data on waste generated, received and treated for 2008–2025, covering about 450 treatment sites per year. For 2023, the register contains 4,102 treatment records from 419 sites and 31,801 waste generation records.

Data reported to ROO for the Gospić site:

Year	Waste treated (t)	Waste generated (t)
2022	549	549
2023	15,574 (R3, almost all code 19 12 04, plastics)	19
2024–2025	no report	no report

Input rose about 28 times in one year, reported residue was 0.1% of input, and reporting then stopped. The difference of about 15,500 t must be either product or stock, and neither is included in the public data.

The analysis also showed that a simple output/input ratio is not sufficient on its own. In 2023, 182 sites treated at least 1,000 t with a process other than storage or landfill. Of these, 86 reported output below 5% of input. Most of them are likely legitimate recyclers, because ROO does not record products that are no longer classified as waste. Within this group, the Gospić operator is 31st by input. Process models and independent data sources are therefore necessary to separate real deviations from normal recycling.

5. Related work

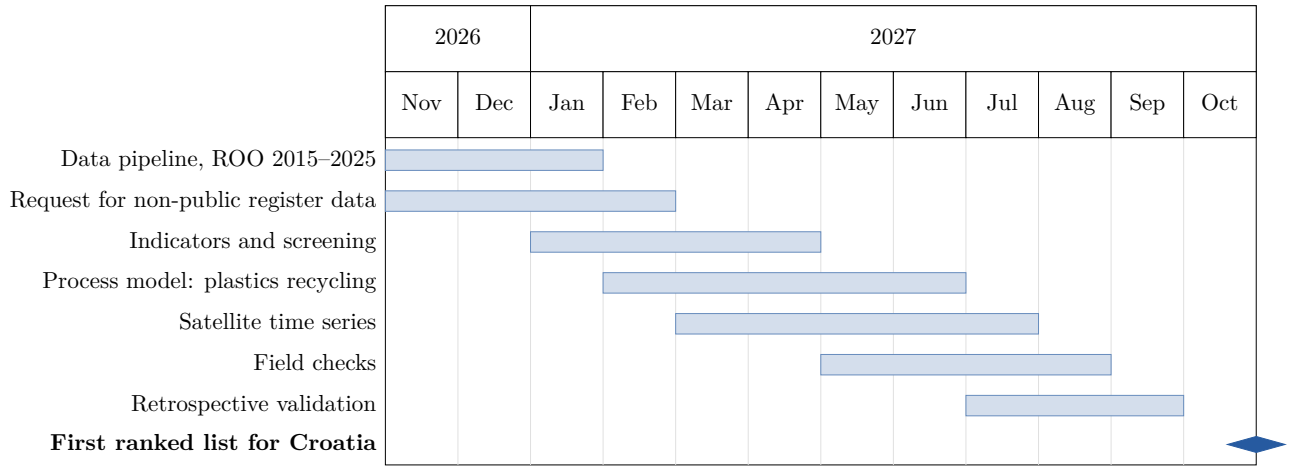
Electronic consignment notes (e-ONTO in Croatia, DIWASS for cross-border shipments in the EU since May 2026) record each waste transfer but do not check whether the reported quantities are consistent. Satellite detection of illegal dumps is being developed, for example in the PERIVALLON project at Politecnico di Milano and by commercial earth-observation companies, but it does not link a dump to a specific facility's reports. Material flow analysis software such as STAN (TU Wien) is used in research for one-off studies. We have not found a tool that combines facility-level mass balance, process and chemical models, and independent data sources for continuous monitoring.

6. Team

Member	Field	Role in the project
Marin Gracin	Chemical engineering, Faculty of Chemical Engineering and Technology	Process yields, chemical feasibility, element balances (chlorine, heavy metals)
Roko Rukavina	Mechanical engineering, Faculty of Mechanical Engineering and Naval Architecture	Plant capacity models, energy use per tonne, stockpile volume from drone and satellite data
Bartol Gracin	Mathematics and programming, Faculty of Science	Data pipeline, data reconciliation, partner network analysis, ranking

7. Work plan

The planned duration is 12 months, from November 2026 to October 2027.



The request for non-public data (stock levels, transfers between partners, exports) is sent at the start, since the answer affects the screening and the validation. The ranking is checked against known cases, such as revoked permits, inspection findings and court decisions, before any results are shared.

8. Limitations and open questions

- Public ROO data does not include stock levels, recipients of waste, or products after end-of-waste status. Without them, many legitimate recyclers will appear as deviations.
- Annual data becomes public with a delay of about 15 months. Satellite images and permit history may give earlier signals.
- If partners coordinate their false reports, the mass balance can close on paper. Only independent sources can detect this.
- Waste moved to an unregistered third location does not appear in any balance.
- The number of confirmed cases available for validation is small.
- The terms of use for bulk access to ROO data still need to be confirmed with the register operator.

9. Next steps

1. Submit a request for access to information to the Ministry for full 2019–2025 records.
2. Confirm the terms of use and bulk data access with the ROO operator.
3. Load public ROO data for 2015–2025 into a local database.
4. Assemble a set of known cases for validation.